Markup Accounting

Juan Holguín
UWO

Sergio Ocampo
UWO

Sergio Salgado
Wharton

May, 2025

Accounting for markup dispersion

- Markups are dispersed across firms, even within narrowly defined markets
- Market concentration $\longrightarrow$ Market power $\longrightarrow$ Link market shares and markups
  - Markup dispersion linked to misallocation and welfare losses

    (Baqaee, Farhi, Sangani, 2023; Edmond, Midrigan, Xu, 2023; Albrecht, Phelan, Pretnar, 2023; Dhingra, Morrow 2019; Zhelobodko, Kokovin, Parenti, and Thisse, 2012; Dixit, Stiglitz, 1977)

Accounting for markup dispersion

- Markups are dispersed across firms, even within narrowly defined markets
- Market concentration $\longrightarrow$ Market power $\longrightarrow$ Link market shares and markups
    - Markup dispersion linked to misallocation and welfare losses

      (Baqaee, Farhi, Sangani, 2023; Edmond, Midrigan, Xu, 2023; Albrecht, Phelan, Pretnar, 2023; Dhingra, Morrow 2019; Zhelobodko, Kokovin, Parenti, and Thisse, 2012; Dixit, Stiglitz, 1977)

This paper:

1. New facts on markup dispersion and market concentration

Accounting for markup dispersion

- Markups are dispersed across firms, even within narrowly defined markets
- Market concentration $\longrightarrow$ Market power $\longrightarrow$ Link market shares and markups
  - Markup dispersion linked to misallocation and welfare losses

    (Baqaee, Farhi, Sangani, 2023; Edmond, Midrigan, Xu, 2023; Albrecht, Phelan, Pretnar, 2023; Dhingra, Morrow 2019; Zhelobodko, Kokovin, Parenti, and Thisse, 2012; Dixit, Stiglitz, 1977)

This paper:

1. New facts on markup dispersion and market concentration
2. Analytical model of oligopolistic competition with flexible non-CES demand

Accounting for markup dispersion

- Markups are dispersed across firms, even within narrowly defined markets
- Market concentration $\longrightarrow$ Market power $\longrightarrow$ Link market shares and markups
  - Markup dispersion linked to misallocation and welfare losses
    (Baqaee, Farhi, Sangani, 2023; Edmond, Midrigan, Xu, 2023; Albrecht, Phelan, Pretnar, 2023; Dhingra, Morrow 2019; Zhelobodko, Kokovin, Parenti, and Thisse, 2012; Dixit, Stiglitz, 1977)

This paper:

1. New facts on markup dispersion and market concentration
2. Analytical model of oligopolistic competition with flexible non-CES demand
3. How much of markup dispersion is accounted by market shares?

Accounting for markup dispersion

- Markups are dispersed across firms, even within narrowly defined markets
- Market concentration $\longrightarrow$ Market power $\longrightarrow$ Link market shares and markups
    - Markup dispersion linked to misallocation and welfare losses

      (Baqaee, Farhi, Sangani, 2023; Edmond, Midrigan, Xu, 2023; Albrecht, Phelan, Pretnar, 2023; Dhingra, Morrow 2019; Zhelobodko, Kokovin, Parenti, and Thisse, 2012; Dixit, Stiglitz, 1977)

This paper:

1. New facts on markup dispersion and market concentration
2. Analytical model of oligopolistic competition with flexible non-CES demand
3. How much of markup dispersion is accounted by market shares?
4. Efficiency losses from different sources of markup dispersion

Markup dispersion concentrated between firms of similar size

- Estimate markups for different countries and periods using different methodologies
    - Data on Indian, Colombian, and U.S. firms
    - Revenue and price-based markup estimates
    - Industry and product level markups

Markup dispersion concentrated between firms of similar size

- Estimate markups for different countries and periods using different methodologies
  - Data on Indian, Colombian, and U.S. firms
  - Revenue and price-based markup estimates
  - Industry and product level markups
- **Over half of markup variance** accounted by dispersion between firms of similar size
  - Within-size component: 57–74% with revenue-based markups
  - Within-size component: 77% with Indian price-based markups

Markup dispersion concentrated between firms of similar size

- Estimate markups for different countries and periods using different methodologies
  - Data on Indian, Colombian, and U.S. firms
  - Revenue and price-based markup estimates
  - Industry and product level markups
- **Over half of markup variance** accounted by dispersion between firms of similar size
  - Within-size component: 57–74% with revenue-based markups
  - Within-size component: 77% with Indian price-based markups
- Accounting for markups requires **high-markup small firms and low-markup large firms**

Markups beyond market concentration

- Model built to account for sales–markup distribution

    1. Large firms internalize market demand (Atkeson, Burstein, 2008)
    2. Demand elasticity varies **flexibly** with firm size (Kimball, 1995; Matsuyama, Ushchev, 2017)

Key: Firm-specific demand elasticity shifters $\longrightarrow$ Deviations from concentration-based forces

Markups beyond market concentration

- Model built to account for sales–markup distribution
    1. Large firms internalize market demand (Atkeson, Burstein, 2008)
    2. Demand elasticity varies **flexibly** with firm size (Kimball, 1995; Matsuyama, Ushchev, 2017)

  Key: Firm-specific demand elasticity shifters $\longrightarrow$ Deviations from concentration-based forces

- Without elasticity shifters $\longrightarrow$ Miss 80% of markup dispersion
    - Generate counterfactual markup-concentration relationship $\longrightarrow$ 2/3 between firm size

Markups beyond market concentration

- Model built to account for sales–markup distribution

  1. Large firms internalize market demand (Atkeson, Burstein, 2008)
  2. Demand elasticity varies **flexibly** with firm size (Kimball, 1995; Matsuyama, Ushchev, 2017)

  Key: Firm-specific demand elasticity shifters $\longrightarrow$ Deviations from concentration-based forces

- Without elasticity shifters $\longrightarrow$ Miss 80% of markup dispersion

  - Generate counterfactual markup-concentration relationship $\longrightarrow$ 2/3 between firm size

- Accounting for markups between similarly sized firms matters

  - Misallocation costs: *Distance to the frontier* 28% (Baqaee, Farhi, 2020)
  - Markup dispersion tied to dispersion in substitutability (Different from "*Darwinian*" logic)

Data and Markup Measurement

Data sources and definitions

Country	Source	Data	Period	Markup Estimates
India	Annual Survey of Industries	Price + Quantity	2001–2008	Raval 2023 DLGKP 2016 + ACF 2015
Colombia	Encuesta Anual Manufacturera	Revenue	1980–1989	Raval 2023
United States	Compustat	Revenue	1980–1989	DLEU 2020

Markets:

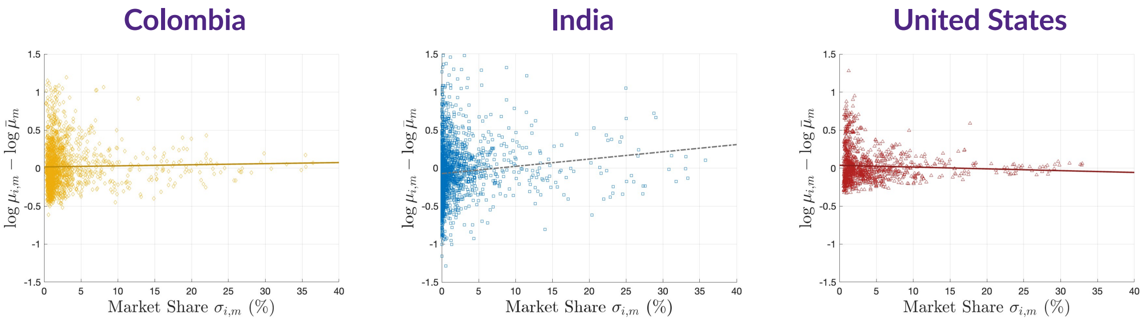
▸ examples

- Industry markets in Colombia and US; Product markets in India (Smith, Ocampo, 2025)
- Firm size is market revenue share, σ_i^m

Markups: Production function estimation and cost minimization

▸ details

Markup dispersion and market concentration



- Firm markups (μ_i^m) relative to market markup $(\overline{\mu}_m)$
- Markup dispersion is large among small firms
- Large firms often have below-average markups

Variance decomposition: Dispersion between and within firm size

- Divide firms in each market by market share in 5pp bins
- Decompose market variance in markups $\longrightarrow$ Report average shares by country

Variance decomposition: Dispersion between and within firm size

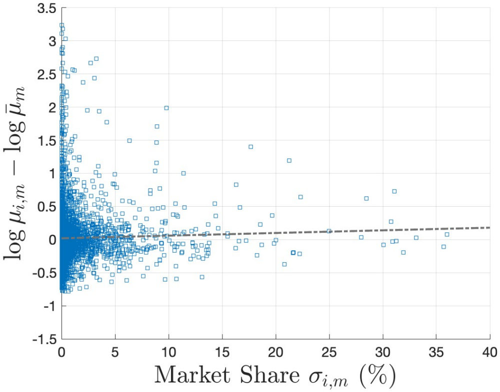
- Divide firms in each market by market share in 5pp bins
- Decompose market variance in markups $\longrightarrow$ Report average shares by country

	Variance	Share within	Share between	Corr($\log \mu_i, \sigma_i$)
Colombia	0.059	0.74	0.26	0.03
India	0.231	0.54	0.46	0.14
United States	0.024	0.68	0.32	-0.08

- Within-bin variation accounts for at least half of markup variance
- The markup-size correlation is weak

Dispersion of price-based markups

▸ details



- **73%** of markup variance within 5pp market share bins (53% in 1pp bins)
- Variance of price-markups smaller than revenue-markups
- Robust to time aggregation, single product firms, and more ▸ robustness

▸ Quant. Shares ▸ log. Shares ▸ State Shares ▸ Age vs μ

Oligopolistic Competition with Idiosyncratic Demands

Model objectives and overview

O1: Match distribution of markups and firm size

$\longrightarrow$ Small firms with high markups + Large firms with low markups

O2: Disentangle role of heterogeneity in productivity, demand, and market concentration

Model objectives and overview

O1: Match distribution of markups and firm size

$\longrightarrow$ Small firms with high markups + Large firms with low markups

O2: Disentangle role of heterogeneity in productivity, demand, and market concentration

Model Ingredients

Oligopolistic Competition	+	Variable Elasticity of Demand	+	Productivity & Demand Shifters

Model objectives and overview

O1: Match distribution of markups and firm size

$\longrightarrow$ Small firms with high markups + Large firms with low markups

O2: Disentangle role of heterogeneity in productivity, demand, and market concentration

Model Ingredients

Oligopolistic Competition	+	Variable Elasticity of Demand	+	Productivity & Demand Shifters

- Demand for market's goods from CES aggregator: $\frac{P_m}{P} = \alpha_m \left(\frac{Y_m}{Y} \right)^{-\frac{1}{\gamma}}$
- Homothetic Direct Implicit Additivity Aggregator for market output (Matsuyama 2023)

Equilibrium markups (Cournot)

▸ How? ▸ Bertrand ▸ Aggregation

$$\frac{1}{\mu_i^m} = \underbrace{\frac{\gamma-1}{\gamma}}_{\text{Monopoly Markup}} + \underbrace{\left(\frac{1}{\gamma} - \frac{1}{\varepsilon_i^m}\right)\left(1-\sigma_i^m\right)}_{\text{“}i\text{” vs Market}} + \underbrace{\left(\frac{1}{\varepsilon_i^m} - E_\sigma\left[\frac{1}{\varepsilon_j^m}\right]\right)\sigma_i^m}_{\text{“}i\text{” vs Competitors “}j\text{”}}$$

Equilibrium markups (Cournot)

▸ How? ▸ Bertrand ▸ Aggregation

$$\frac{1}{\mu_i^m} = \underbrace{\frac{\gamma - 1}{\gamma}}_{\text{Monopoly Markup}} + \underbrace{\left(\frac{1}{\gamma} - \frac{1}{\varepsilon_i^m}\right)(1 - \sigma_i^m)}_{\text{“}i\text{” vs Market}} + \underbrace{\left(\frac{1}{\varepsilon_i^m} - E_\sigma\left[\frac{1}{\varepsilon_j^m}\right]\right)\sigma_i^m}_{\text{“}i\text{” vs Competitors “}j\text{”}}$$

Higher markup μ_i^m if

- “Own elasticity” (ε_i^m) lower than market’s (γ)
- “Own variety” is elastic relative to market average (limiting substitution effects)

Limit cases: When $\varepsilon_i^m = \nu \longrightarrow$ Atkeson & Burstein; When $\sigma_i^m = 0 \longrightarrow$ Kimball

Estimation and Model Objects

Indian product markets with price-based markups

Recovering own elasticities $\{\varepsilon_i^m\}$

▸ ε distribution ▸ inversion

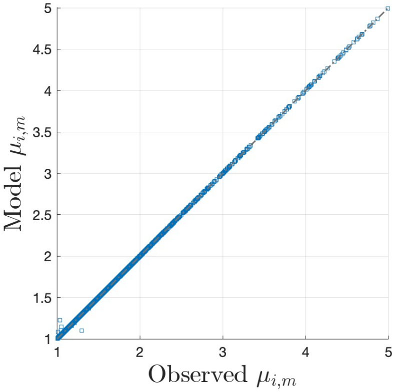
1. Observe markups and market shares $\{\mu_i^m, \sigma_i^m\}$

2. Set market elasticity $\gamma = 1.3$

   (Edmond, Midrigan, Xu, 2023)

3. Invert markups to recover $\{\varepsilon_i^m\}$

   This does not require parametric form for Υ_i



Estimating demand parameters: Elasticity shifters

Adopt parametric for Υ from from Klenow, Willis (2016)

$$\text{Elasticity:} \quad \varepsilon_i^m = \nu_i^m \left(\frac{y_i^m}{Y_m}\right)^{-\theta_m/\nu_i^m}, \qquad \text{Super-elasticity:} \quad \xi_i^m = \theta_m \left(\frac{y_i^m}{Y_m}\right)^{-\theta_m/\nu_i^m}$$

Elasticity shifters: ν_i^m Super-elasticity market parameter: θ_m

Estimating demand parameters: Elasticity shifters

Adopt parametric for Υ from from Klenow, Willis (2016)

$$\text{Elasticity:} \quad \varepsilon_i^m = \nu_i^m \left(\frac{y_i^m}{Y_m}\right)^{-\theta_m/\nu_i^m}, \qquad \text{Super-elasticity:} \quad \xi_i^m = \theta_m \left(\frac{y_i^m}{Y_m}\right)^{-\theta_m/\nu_i^m}$$

Elasticity shifters: ν_i^m Super-elasticity market parameter: θ_m

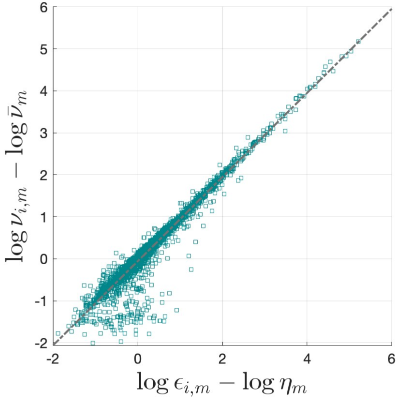
Changes in elasticity and market shares:

$$d\log\varepsilon_i^m = -\left(\frac{\xi_i^m}{\varepsilon_i^m}\right)\left(\frac{\varepsilon_i^m}{\varepsilon_i^m - 1}\right) d\log\sigma_i^m = -\left(\frac{\theta^m}{\nu_i^m}\right)\left(\frac{\varepsilon_i^m}{\varepsilon_i^m - 1}\right) d\log\sigma_i^m$$

- Estimate θ_m and $\{\nu_i^m\}$ by method of moments

Elasticity and elasticity shifters

Shifters $\{\nu_i^m\}$ and own-elasticities $\{\varepsilon_i^m\}$

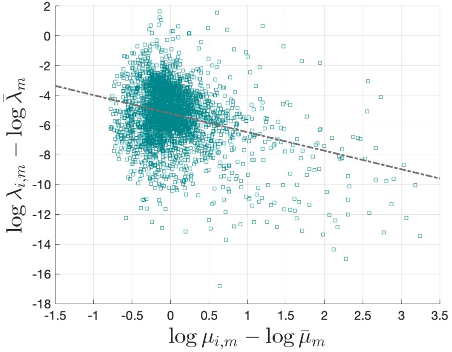


Much of variation in the own-elasticities $\{\varepsilon\}$ is not explained by firm size

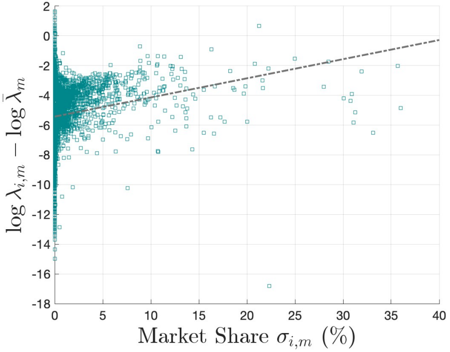
Marginal costs, markups, and firm size

Recovering model objects

Marginal costs (λ) and markups (μ)



Marginal costs (λ) and market share (σ)



- High-markup firms tend to have lower marginal costs
- Larger firms tend to have higher marginal costs on average

Market Structure, Firm Scale, and Demand Factors

Alternative models of variable markups

Objective: Disentangle role of market concentration, scale, and demand factors

- Construct counterfactual markups under alternative models

Model	Strategic interaction	Variable elasticity	Firm-specific shifter
Atkeson-Burstein	Yes	No	No
Kimball	No	Yes	No
A&B + Kimball	Yes	Yes	No
Full model	Yes	Yes	Yes

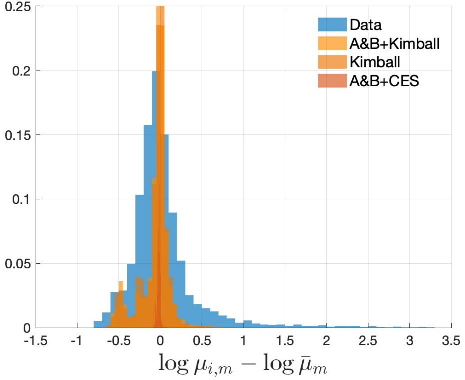
- All alternatives match market shares and average market markups

- We look at the dispersion of markups and their distribution over firm size

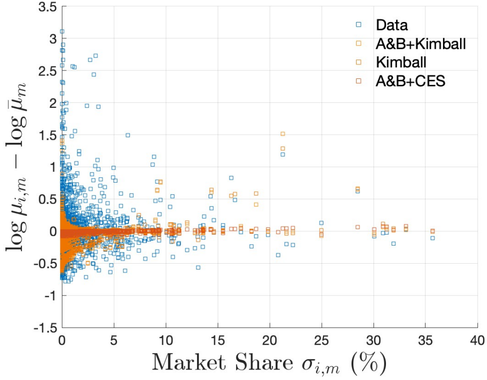
Markup distributions across models

▸ Col/US

Markup distribution



Markup and market share



Models without elasticity shifters miss small high-markup firms and large low-markup firms

Variance decomposition across models

Col/US

	Variance	Share within	Share between	Corr(log μ_i, σ_i)
Data	0.196	0.77	0.23	0.06
Full model	0.195	0.77	0.23	0.06
A&B + Kimball	0.047	0.39	0.61	0.21
Kimball	0.048	0.34	0.66	0.22
Atkeson-Burstein	0.001	0.18	0.83	0.34

- Alternative models generate too little dispersion
- They also assign too much variation to differences across firm sizes (market share)

Markups and misallocation

- Compute distance to the efficient frontier $\left(\frac{Y}{Y^{\star}}\right)$ following Baqaee & Farhi (2020)
  - Incorporate non-CES into allocation patterns

$$\varphi_{ij}^{m} \equiv \frac{1}{\sigma_j^m} \frac{\partial \log y_j^m}{\partial \log p_i^m}\bigg|_{Y_m} = \begin{cases} \frac{\varepsilon_i^m \varepsilon_j^m}{E_\sigma[\varepsilon_h]} & \text{if } j \neq i \\ -\frac{\varepsilon_i^m}{\sigma_i^m}\left(1 - \frac{\sigma_i^m \varepsilon_i^m}{E_\sigma[\varepsilon_h]}\right) & \text{if } j = i \end{cases}$$

Markups and misallocation

- Compute distance to the efficient frontier $\left(\frac{Y}{Y^{\star}}\right)$ following Baqaee & Farhi (2020)
    - Incorporate non-CES into allocation patterns

$$\varphi_{ij}^{m} \equiv \frac{1}{\sigma_j^m} \frac{\partial \log y_j^m}{\partial \log p_i^m}\bigg|_{Y_m} = \begin{cases} \frac{\varepsilon_i^m \varepsilon_j^m}{E_\sigma[\varepsilon_h]} & \text{if } j \neq i \\ -\frac{\varepsilon_i^m}{\sigma_i^m}\left(1 - \frac{\sigma_i^m \varepsilon_i^m}{E_\sigma[\varepsilon_h]}\right) & \text{if } j = i \end{cases}$$

- Dispersion in own-elasticities (and elasticity shifters) affects reallocation in two ways:

(i) Reflect substitutability **(ii)** Generate productive small firms

Markups and misallocation

- Compute distance to the efficient frontier $\left(\frac{Y}{Y^\star}\right)$ following Baqaee & Farhi (2020)
  - Incorporate non-CES into allocation patterns

$$\varphi_{ij}^m \equiv \frac{1}{\sigma_j^m} \frac{\partial \log y_j^m}{\partial \log p_i^m}\bigg|_{Y_m} = \begin{cases} \frac{\varepsilon_i^m \varepsilon_j^m}{E_\sigma[\varepsilon_h]} & \text{if } j \neq i \\ -\frac{\varepsilon_i^m}{\sigma_i^m}\left(1 - \frac{\sigma_i^m \varepsilon_i^m}{E_\sigma[\varepsilon_h]}\right) & \text{if } j = i \end{cases}$$

- Dispersion in own-elasticities (and elasticity shifters) affects reallocation in two ways:

(i) Reflect substitutability **(ii)** Generate productive small firms

	$\log(Y^\star/Y)$
Full model	0.25
A&B + Kimball	0.05
Kimball	0.05
Atkeson-Burstein	0.02

- Output loss of 28%

  (comparable with EMX)

- 5 times larger than loss without elasticity shifters

Conclusions

- Most markup dispersion is concentrated in firms of similar size
- Accounting for markups (and market concentration!) reveals role of **demand factors**
  - Active literature on this front:

    Blum, Claro, Horstmann, Rivers, 2024; Eslava, Haltiwanger, Urdaneta, 2023; Afrouzi, Drenik, Kim, 2025; Hubmer, Nord, 2025; Casal, 2026; Haddara, 2026; and more!
- Demand factors weaken link between firm size, productivity, and market power
- Ignoring demand heterogeneity understates the efficiency cost of markup dispersion

Appendix

Appendix: firms, sectors, and products

back

Indian ASI product markets

Sector	Example products	Product revenue share in sector
Bakery products	Biscuits & cookies; bread, buns & croissants	0.83; 0.10
Pesticides & insecticides	Pesticides; insecticides	0.59; 0.23
Soaps & detergents	Detergent powder; toilet soap	0.20; 0.34
Leather footwear	Boots; leather shoe uppers	0.50; 0.25
Audio/video apparatus	Printed circuit boards; color TV sets	0.06; 0.61

- India: product-year markets; production functions estimated at the 3-digit sector level.
- Colombia and U.S.: industry-year markets; prices and quantities are not separately observed.
- Multi-product establishments are assigned product-line inputs using within-firm revenue shares.

Appendix: Measuring markups from production functions

back

Cost minimization: Firm i in market m (independent of demand-side)

$$C_i^m(y) = \min_{\{x_n\}} \sum_{n=1}^{N} p_n x_n \qquad \text{s.t.} \qquad y \leq z_i^m F_m(x_1, \ldots, x_N, K_1, \ldots, K_H)$$

Markup formula

$$\mu_i \equiv \frac{p_i}{\lambda_i} = \frac{\epsilon_{x_n,i}}{\omega_{x_n,i}} \begin{array}{l} \rightarrow \text{Output elasticity to } x_n \\ \rightarrow \text{Revenue share of } x_n \end{array}$$

- Input shares $\{\omega\}$ are observed from revenue data
- Output elasticities $\{\epsilon\}$ from production-function estimation
  (DLEU 2020; Raval 2023; DLGKP 2016; ACF 2015)

Details

Appendix: revenue-based production-function approach

back

Cost-share approach

- Used for Colombia, U.S., and revenue-based Indian robustness.
- Relies on cost minimization and constant returns to scale in observed flexible inputs.
- Recovers output elasticities from average input cost shares.

$$E[P_n X_{n,i}] = \epsilon_n E[\lambda_i Y_i], \qquad \epsilon_n = \frac{E[P_n X_{n,i}]}{E[\sum_{k=1}^{N} P_k X_{k,i}]}$$

Appendix: Price-based product markups

back

- Revenue-based markups are widely available but subject to output-price bias

  (Bond, Hashemi, Kaplan, Zoch, 2021)

- Quantity-based production functions estimated from price-quantity data address this
- Estimate for India at the 3-digit sector level using single-product firms
  - Bakery products; Soaps and detergents; Leather footwear; etc.
- Multi-product firm inputs allocated using within-firm revenue shares

  (results robust to single-product firms)

Details

Appendix: quantity-based production-function approach

Indian ASI price and quantity data

$$q_{ijt} = f(m_{ijt}, l_{ijt}, k_{ijt}) + \omega_{ijt} + \zeta_{ijt}$$

- Quantity production functions remove output-price bias.
- Estimated at the 3-digit sector level.
- Single-product firms identify production parameters without product-level input allocation.
- Multi-product firms receive sector elasticities and product-level input allocations.
- Implemented with Wooldridge's control-function estimator; first-order lags instrument materials, output prices, and market shares.

Appendix: details on quantity-production estimation

◂ back

- Product-level input allocations are unobserved for multi-product firms.
- Estimation uses single-product firms, so $\rho_{x,ijt} = 1$.
- Output prices help proxy for unobserved input quality.
- State fixed effects absorb geographic input-price differences.
- Baseline assumes capital and labor are predetermined, materials are flexible.

Markup at product level

$$\mu_{ijt} = \epsilon_{x,ijt} \frac{P_{ijt} Q_{ijt}}{\rho_{x,ijt} R_{x,it}}$$

Appendix: price-based markup robustness

back

Observed price-based markup variance: India, 2001–2008

	Baseline	Avg. across years	At least 2	At least 3	At least 4
Variance	0.196	0.232	0.177	0.169	0.114
% Within (1pp)	0.564	0.833	0.705	0.550	0.399
% Within (5pp)	0.770	0.961	0.860	0.783	0.693
Avg. number firms	386	386	142	66	32

- Columns restrict to firms observed producing the same product for more years.

- Within-size dispersion remains large in the more persistent-product samples.

Single product firms

	Variance	Share within	Share between	Corr($\log \mu_i, \sigma_i$)
All firms	0.196	0.770	0.230	0.062
Single product	0.183	0.506	0.494	0.071

Appendix: quantity-based markup decomposition

back

Markup Variance Decomposition India (2001-2008) - Quantity based market shares

	Variance	Share within	Share between	Corr(log μ_i, σ_i)
Data	0.200	0.716	0.284	0.034
Full Model	0.199	0.718	0.282	0.045
A&B + Kimball	0.026	0.546	0.454	0.097
Kimball	0.022	0.508	0.492	0.099
Atkeson & Burstein	0.001	0.386	0.614	0.170

Market *HDIA* aggregator

$$1 = \sum_{i=1}^{N_m} \Upsilon_i^m \left(\frac{y_i^m}{Y_m} \right) \qquad \text{CES: } \Upsilon\left(\frac{y_i^m}{Y_m} \right) = \left(\frac{y_i^m}{Y_m} \right)^{\frac{\nu-1}{\nu}}$$

Market *HDIA* aggregator

$$1 = \sum_{i=1}^{N_m} \Upsilon_i^m \left(\frac{y_i^m}{Y_m} \right) \qquad \text{CES: } \Upsilon \left(\frac{y_i^m}{Y_m} \right) = \left(\frac{y_i^m}{Y_m} \right)^{\frac{\nu-1}{\nu}}$$

Inverse demand for firm *i*

$$\frac{p_i^m}{P_m} = \frac{\left(\Upsilon_i^m\right)' \left(\frac{y_i^m}{Y_m}\right)}{\sum_j \left(\Upsilon_j^m\right)' \left(\frac{y_j^m}{Y_m}\right) \frac{y_j^s}{Y_m}} \qquad \text{CES: } \frac{p_i^m}{P_m} = \left(\frac{y_i^m}{Y_m} \right)^{\frac{-1}{\nu}}$$

Market *HDIA* aggregator

$$1 = \sum_{i=1}^{N_m} \Upsilon_i^m \left(\frac{y_i^m}{Y_m}\right) \qquad \text{CES: } \Upsilon\left(\frac{y_i^m}{Y_m}\right) = \left(\frac{y_i^m}{Y_m}\right)^{\frac{\nu-1}{\nu}}$$

Inverse demand for firm *i*

$$\frac{p_i^m}{P_m} = \frac{\left(\Upsilon_i^m\right)'\left(\frac{y_i^m}{Y_m}\right)}{\sum_j \left(\Upsilon_j^m\right)'\left(\frac{y_j^m}{Y_m}\right)\frac{y_j^s}{Y_m}} \qquad \text{CES: } \frac{p_i^m}{P_m} = \left(\frac{y_i^m}{Y_m}\right)^{\frac{-1}{\nu}}$$

Profit maximization

$$\max \quad p_i^m y_i^m - C_i(y_i^m) \quad \longrightarrow \quad p_i^m = \frac{1}{1-\frac{1}{\eta_i^m}} C_i'\left(y_i^m\right) = \boldsymbol{\mu_i^m} C_i'\left(y_i^m\right)$$

Equilibrium elasticity (Cournot)

back

$$\frac{1}{\eta_i^m} \equiv -\frac{\partial \log p_i^m}{\partial \log y_i^m}$$

Equilibrium elasticity (Cournot)

back

$$\frac{1}{\eta_i^m} = \underbrace{\left(- \frac{\partial \log \Upsilon_i' \left(\frac{y_i^m}{Y_m} \right)}{\partial \log y_i^m} \right)}_{\varepsilon_i^m\text{:Own-Demand Elasticity}}$$

Own elasticity: Captures elasticity over residual demand *all else equal*

$$\varepsilon_i^m = - \frac{\Upsilon_i' \left(\frac{y_i^m}{Y_m} \right)}{\frac{y_i^m}{Y_m} \Upsilon_i'' \left(\frac{y_i^m}{Y_m} \right)}$$

CES: $\varepsilon_i^m = \nu$

Equilibrium elasticity (Cournot)

back

$$\frac{1}{\eta_i^m} = \underbrace{\left(-\frac{\partial \log \Upsilon_i'\left(\frac{y_i^m}{Y_m}\right)}{\partial \log y_i^m}\right)}_{\varepsilon_i^m\text{:Own-Demand Elasticity}} + \underbrace{\left(-\frac{\partial \log \left(\frac{Y_m}{Y}\right)^{-\frac{1}{\gamma}}}{\partial \log y_i^m}\right)}_{\text{Granular Effect}}$$

Own elasticity: Captures elasticity over residual demand *all else equal*

$$\varepsilon_i^m = -\frac{\Upsilon_i'\left(\frac{y_i^m}{Y_m}\right)}{\frac{y_i^m}{Y_m}\Upsilon_i''\left(\frac{y_i^m}{Y_m}\right)} \qquad \text{CES: } \varepsilon_i^m = \nu$$

Equilibrium elasticity (Cournot)

back

$$\frac{1}{\eta_i^m} = \underbrace{\left(-\frac{\partial \log \Upsilon_i'\left(\frac{y_i^m}{Y_m}\right)}{\partial \log y_i^m}\right)}_{\varepsilon_i^m\text{:Own-Demand Elasticity}} + \underbrace{\left(-\frac{\partial \log \left(\frac{Y_m}{Y}\right)^{-\frac{1}{\gamma}}}{\partial \log y_i^m}\right)}_{\text{Granular Effect}} - \underbrace{\left(\frac{\partial \log \sum_j \Upsilon_j'\left(\frac{y_j^m}{Y_m}\right)\frac{y_j^m}{Y_m}}{\partial \log y_i^m}\right)}_{\text{Substitution Effects}}$$

Own elasticity: Captures elasticity over residual demand *all else equal*

$$\varepsilon_i^m = -\frac{\Upsilon_i'\left(\frac{y_i^m}{Y_m}\right)}{\frac{y_i^m}{Y_m}\Upsilon_i''\left(\frac{y_i^m}{Y_m}\right)}$$

CES: $\varepsilon_i^m = \nu$

Equilibrium elasticity (Cournot)

back

$$\frac{1}{\eta_i^m} = \underbrace{\left(-\frac{\partial \log \Upsilon_i'\left(\frac{y_i^m}{Y_m}\right)}{\partial \log y_i^m}\right)}_{\varepsilon_i^m\text{:Own-Demand Elasticity}} + \underbrace{\left(-\frac{\partial \log \left(\frac{Y_m}{Y}\right)^{-\frac{1}{\gamma}}}{\partial \log y_i^m}\right)}_{\text{Granular Effect}} - \underbrace{\left(\frac{\partial \log \sum_j \Upsilon_j'\left(\frac{y_j^m}{Y_m}\right)\frac{y_j^m}{Y_m}}{\partial \log y_i^m}\right)}_{\text{Substitution Effects}}$$

- **Key:** Demand elasticity depends only on own-elasticities and market shares $\{\varepsilon_i^m, \sigma_i^m\}$

Own elasticity: Captures elasticity over residual demand *all else equal*

$$\varepsilon_i^m = -\frac{\Upsilon_i'\left(\frac{y_i^m}{Y_m}\right)}{\frac{y_i^m}{Y_m}\Upsilon_i''\left(\frac{y_i^m}{Y_m}\right)} \qquad \text{CES: } \varepsilon_i^m = \nu$$

Appendix: Bertrand markups

back

$$\frac{1}{\mu_i^m} = 1 - \frac{1}{\gamma\sigma_i^m + \varepsilon_i^m \left[1 - \left(\frac{\varepsilon_i^m}{E_\sigma[\varepsilon_j^m]}\right)\sigma_i^m\right]}$$

- Strategic interaction enters through market share σ_i^m.
- The substitution term depends on own elasticity relative to the market's sales-weighted elasticity.
- This is the baseline formula used for the price-based model exercises.

Appendix: Aggregate markups

back

Market marginal cost

$$\lambda_m = \sum_{i=1}^{N_m} \lambda_i^m \frac{y_i^m}{Y_m}$$

Market markup

$$\mu_m = \frac{P_m}{\lambda_m} = \left[\sum_{i=1}^{N_m} \frac{1}{\mu_i^m} \sigma_i^m \right]^{-1}$$

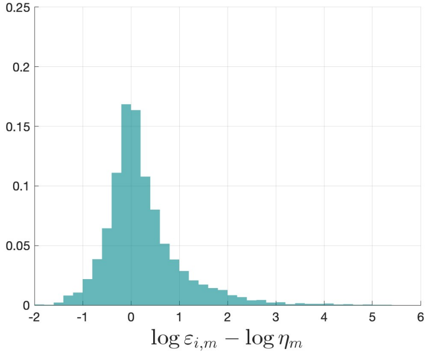
Cournot aggregate expression

$$\frac{1}{\mu_m} = \left(1 - \frac{1}{\gamma}\right) + \left(\frac{1}{\gamma} - E_\sigma\left[\frac{1}{\varepsilon_i^m}\right]\right)(1 - \mathrm{HHI}_m) + 2\,\mathrm{Cov}_\sigma\left(\sigma_i^m, \frac{1}{\varepsilon_i^m}\right)$$

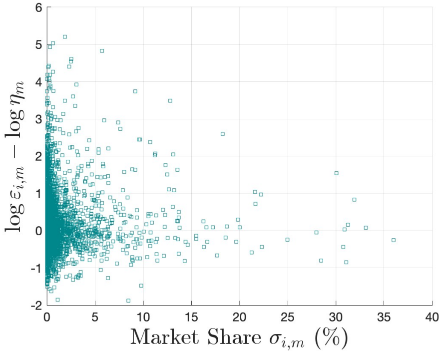
Appendix: Recovered own elasticities

back

Distribution of ε_i^m



ε_i^m and market share

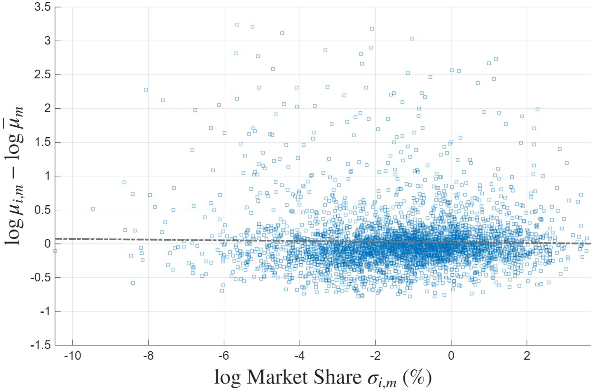


- Own-elasticity dispersion is large, especially among small product lines.
- This is the input into the demand-shifter and model-object recovery steps.

Appendix: Log Market Share vs Markups Scatter

◂ back

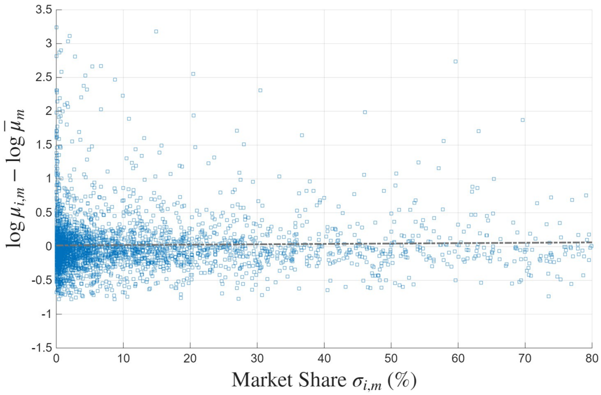
μ_i^m and log market share



Appendix: Within State Market Share vs Markups Scatter

back

μ_i^m and state market share



Appendix: Recovering model objects

back

Recovering relative output

$$\varepsilon_i^m = \nu_i^m \left(\frac{y_i^m}{Y_m}\right)^{-\theta_m/\nu_i^m} \quad \Rightarrow \quad \frac{y_i^m}{Y_m} = \left(\frac{\varepsilon_i^m}{\nu_i^m}\right)^{-\nu_i^m/\theta_m}$$

Recovering relative prices

$$\sigma_i^m = \frac{p_i^m y_i^m}{P_m Y_m} \quad \Rightarrow \quad \frac{p_i^m}{P_m} = \sigma_i^m \frac{Y_m}{y_i^m}$$

- These objects are relative to market-level price and quantity indexes.
- Levels are not needed for the decomposition exercises.

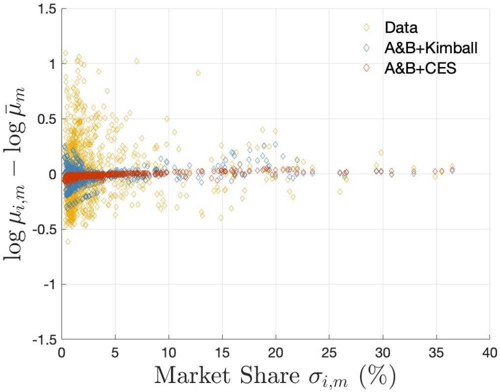
Recovering marginal costs

$$\frac{\lambda_i^m}{\lambda_m} = \frac{\frac{\sigma_i^m}{\mu_i^m}}{\sum_j \frac{\sigma_j^m}{\mu_j^m}} \left(\frac{y_i^m}{Y_m}\right)^{-1}$$

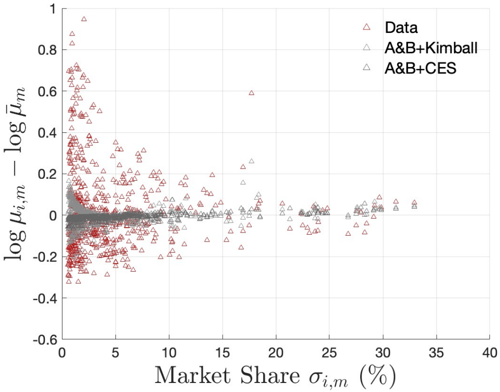
Appendix: markup distributions, Colombia and U.S.

back

Colombia



United States



Model variants shown against the measured markup-size relationship.

Appendix: variance decomposition, Colombia and U.S.

back

Colombia

	Var.	Within	Between	Corr.
Data	0.059	0.524	0.476	0.029
Full model	0.059	0.525	0.475	0.030
Atkeson-Burstein	0.001	0.001	0.999	0.608
Kimball	0.006	0.012	0.988	0.213
A&B + Kimball	0.006	0.021	0.979	0.183

United States

	Var.	Within	Between	Corr.
Data	0.024	0.386	0.614	-0.075
Full model	0.024	0.386	0.614	-0.074
Atkeson-Burstein	0.000	0.001	0.999	0.628
Kimball	0.002	0.008	0.992	-0.018
A&B + Kimball	0.002	0.014	0.986	-0.061

1 percentage point market-share bins. Corr. is Corr($\log \mu_i, \sigma_i$).

Appendix: own-elasticity inversion

back

Cournot inversion is linear in reciprocal elasticities

$$\frac{1}{\varepsilon_i^m}(1-\sigma_i^m) + \sum_{j\neq i} \frac{1}{\varepsilon_j^m}\sigma_j^m \frac{\sigma_i^m}{1-\sigma_i^m} = \frac{1-\frac{1}{\gamma}\sigma_i^m}{1-\sigma_i^m} - \frac{1}{\mu_i^m(1-\sigma_i^m)}$$

Bertrand inversion is nonlinear

$$\frac{\mu_i}{\mu_i - 1} = \gamma\sigma_i + \varepsilon_i\left[1 - \frac{\varepsilon_i\sigma_i}{E_\sigma[\varepsilon]}\right]$$

- Appendix proves uniqueness under the Bertrand markup equation.

Appendix: Relationship Between Age and Markups

back

Table: Firm Age and Markups

	(1) Baseline	(2) With controls	(3) Age < 50
Firm age	0.000	0.001*	0.001
	(0.000)	(0.000)	(0.001)
Log capital-labor ratio		-0.008	-0.008
		(0.006)	(0.006)
Product fixed effects	Yes	Yes	Yes
Year fixed effects	Yes	Yes	Yes
Capital-labor control	No	Yes	Yes
Sample	Full	Full	Age < 50
Observations	9,489	9,454	8,988
Adjusted R^2	0.109	0.110	0.108

Notes: The dependent variable is log markup. Standard errors are reported in parentheses. $^* p < 0.10$.